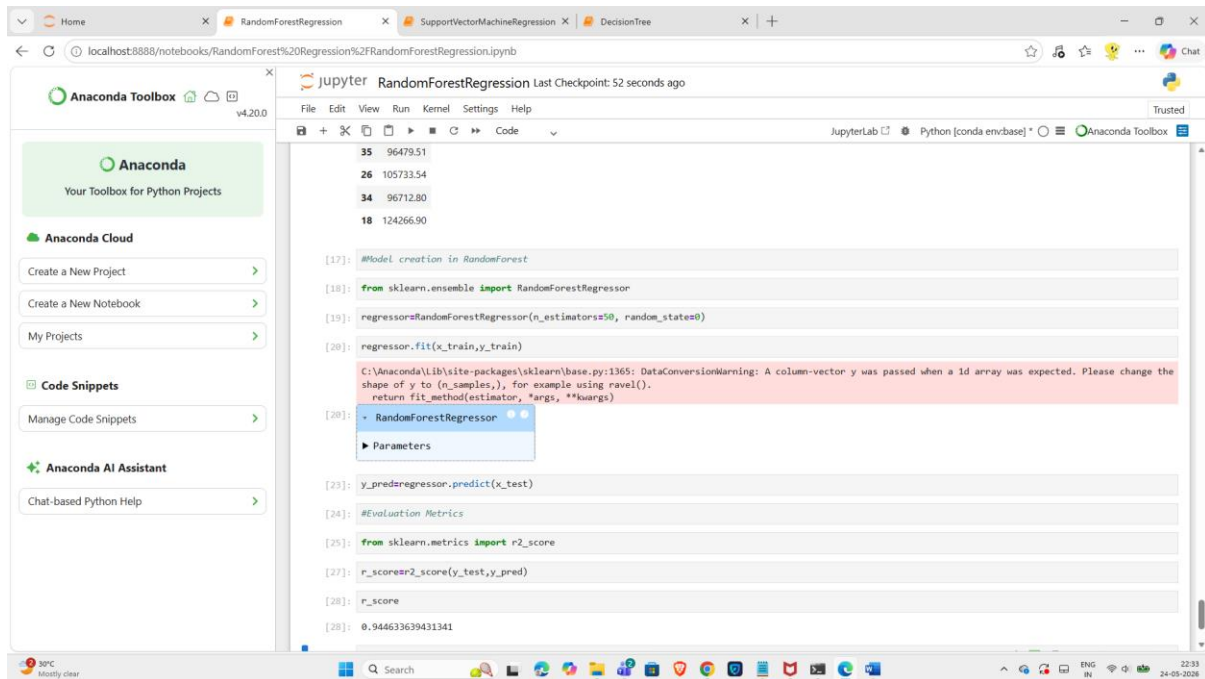


RANDOMFOREST Regression Assignment with different parameters:

`n_estimators=50, random_state=0, r_score=0.944633639431341`

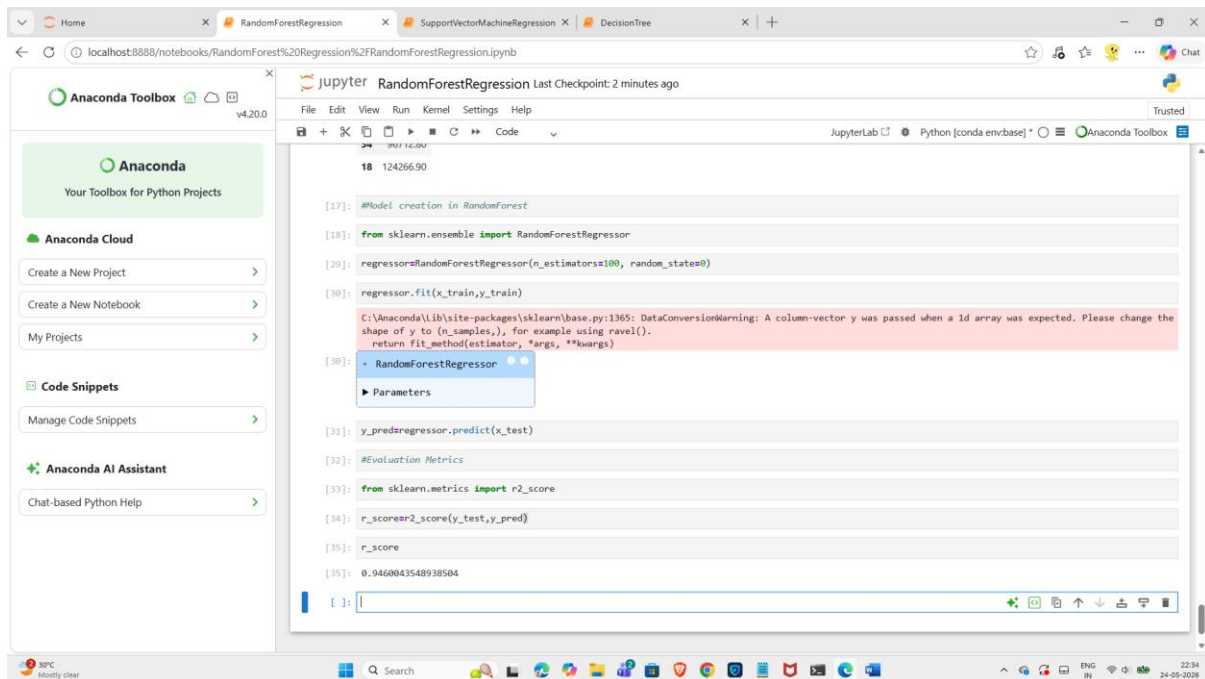


The screenshot shows a Jupyter Notebook interface with the following code and output:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[19]: regressor=RandomForestRegressor(n_estimators=50, random_state=0)
[20]: regressor.fit(x_train,y_train)
[21]: y_pred=regressor.predict(x_test)
[22]: #Evaluation Metrics
[23]: from sklearn.metrics import r2_score
[24]: r_score=r2_score(y_test,y_pred)
[25]: r_score
[26]: 0.944633639431341
```

The output of the notebook shows a prediction of 124266.90 and an r_score of 0.944633639431341. A warning message is displayed: "C:\Anaconda\Lib\site-packages\sklearn\base.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel(). return fit_method(estimator, *args, **kwargs)".

`n_estimators=100, random_state=0, r_score = 0.9460043548938504`



The screenshot shows a Jupyter Notebook interface with the following code and output:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[19]: regressor=RandomForestRegressor(n_estimators=100, random_state=0)
[20]: regressor.fit(x_train,y_train)
[21]: y_pred=regressor.predict(x_test)
[22]: #Evaluation Metrics
[23]: from sklearn.metrics import r2_score
[24]: r_score=r2_score(y_test,y_pred)
[25]: r_score
[26]: 0.9460043548938504
```

The output of the notebook shows a prediction of 124266.90 and an r_score of 0.9460043548938504. A warning message is displayed: "C:\Anaconda\Lib\site-packages\sklearn\base.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel(). return fit_method(estimator, *args, **kwargs)".

S.No	Criterion	Max_features	N_estimators	R_score
1	Absolute_error	sqrt	50	0.7222351871476136
2	Absolute_error	Log2	50	0.7222351871476136
3	Absolute_error	None	50	0.9401935247161504
4	Absolute_error	sqrt	100	0.7857483352154211
5	Absolute_error	Log2	100	0.7857483352154211
6	Absolute_error	None	100	0.9459097460494243
7	Friedman_mse	sqrt	50	0.6889182130535486
8	Friedman_mse	Log2	50	0.6889182130535486
9	Friedman_mse	None	50	0.9388957628188894
10	Friedman_mse	sqrt	100	0.7608592212109025
11	Friedman_mse	Log2	100	0.7608592212109025
12	Friedman_mse	None	100	0.9412701965688807
13	poisson	sqrt	50	0.720862466757838
14	poisson	Log2	50	0.720862466757838
15	poisson	None	50	0.9463549705311108
16	poisson	sqrt	100	0.7717642068103981
17	poisson	Log2	100	0.7717642068103981
18	poisson	None	100	0.9413889418631131

BEST MODEL Highlighted above –

Deploy Code is written for the same and shared in GitHub

Absolute_error	sqrt	50
----------------	------	----

The screenshot shows a Jupyter Notebook titled 'RandomForestRegression' with the following code and output:

```
[26]: 105733.54
[34]: 96712.80
[18]: 124266.90

[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[43]: regressor=RandomForestRegressor(n_estimators=50, criteria='absolute_error', random_state=0, max_features='sqrt')
[44]: regressor.fit(x_train,y_train)
C:\Anaconda\lib\site-packages\sklearn\base.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
  return fit_method(estimator, *args, **kwargs)
[44]: RandomForestRegressor
Parameters
[45]: y_pred=regressor.predict(x_test)
[46]: #Evaluation Metrics
[47]: from sklearn.metrics import r2_score
[48]: r_score=r2_score(y_test,y_pred)
[49]: r_score
[49]: 0.7222351871476136
```

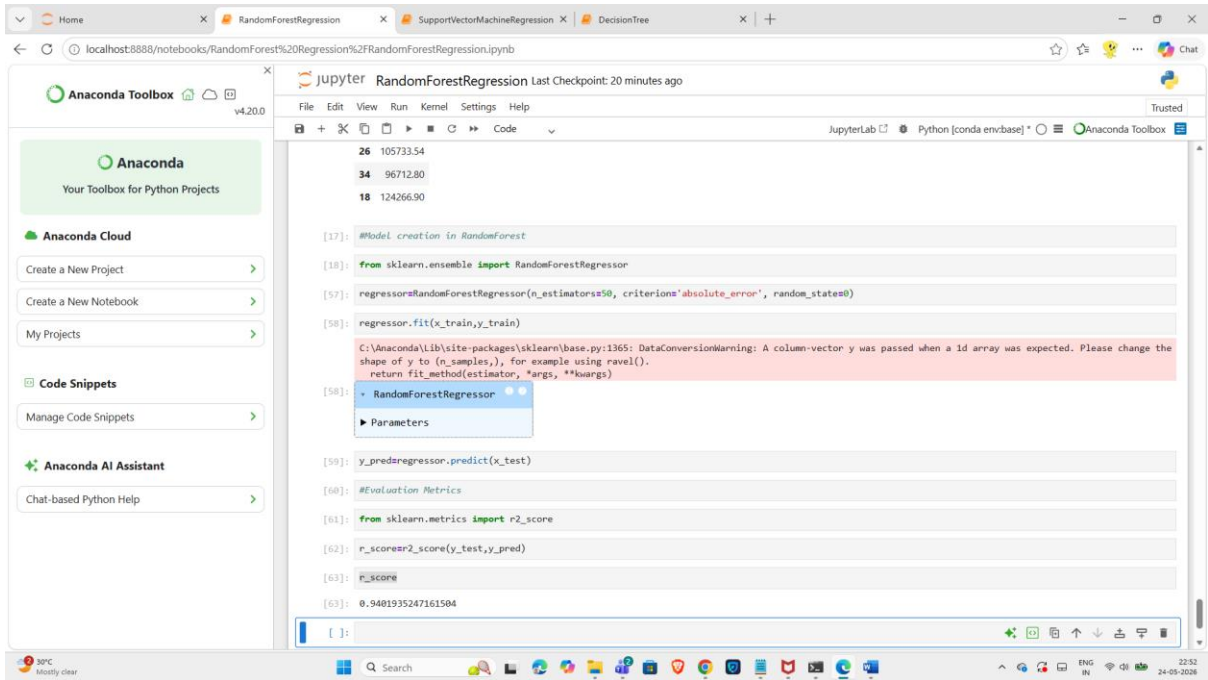
Absolute_error	Log2	50
----------------	------	----

The screenshot shows a Jupyter Notebook titled 'RandomForestRegression' with the following code and output:

```
[26]: 105733.54
[34]: 96712.80
[18]: 124266.90

[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[50]: regressor=RandomForestRegressor(n_estimators=50, criteria='absolute_error', random_state=0, max_features='log2')
[51]: regressor.fit(x_train,y_train)
C:\Anaconda\lib\site-packages\sklearn\base.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
  return fit_method(estimator, *args, **kwargs)
[51]: RandomForestRegressor
Parameters
[52]: y_pred=regressor.predict(x_test)
[53]: #Evaluation Metrics
[54]: from sklearn.metrics import r2_score
[55]: r_score=r2_score(y_test,y_pred)
[56]: r_score
[56]: 0.7222351871476136
```

Absolute_error	None	50
----------------	------	----

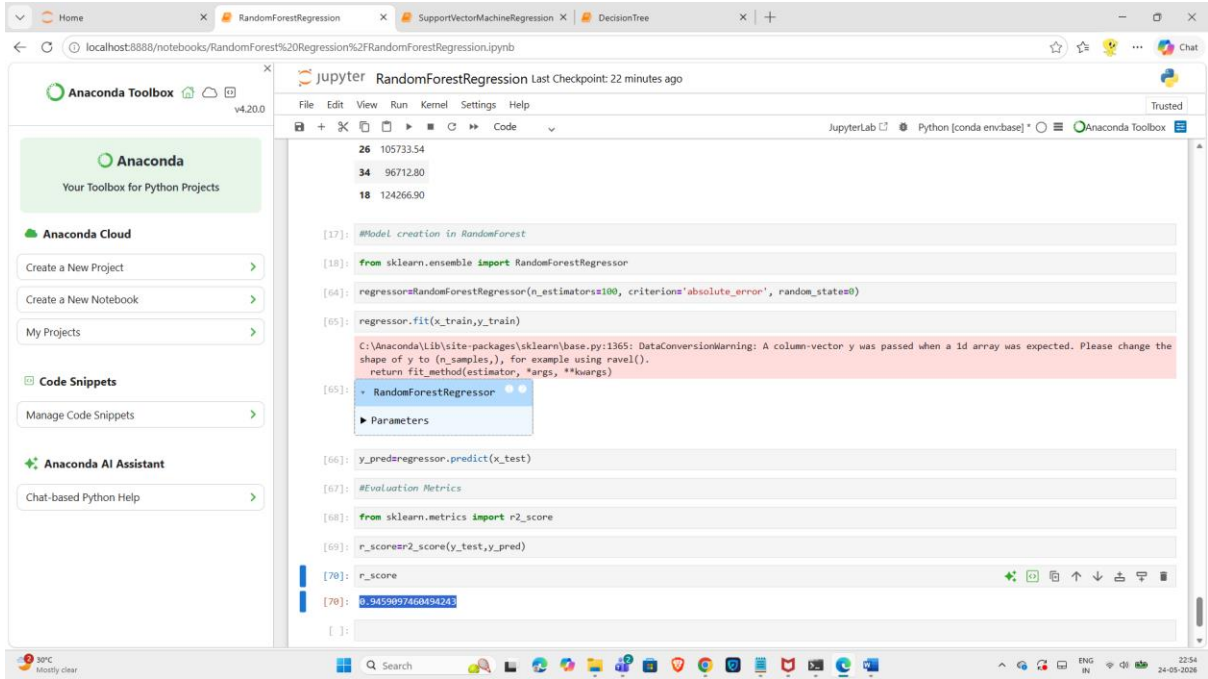


The screenshot shows a Jupyter Notebook interface with the following code and output:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[57]: regressor=RandomForestRegressor(n_estimators=50, criterion='absolute_error', random_state=0)
[58]: regressor.fit(x_train,y_train)
[59]: y_pred=regressor.predict(x_test)
[60]: #Evaluation Metrics
[61]: from sklearn.metrics import r2_score
[62]: r_score=r2_score(y_test,y_pred)
[63]: r_score
[63]: 0.9481935247161504
```

A warning message is displayed: `C:\Anaconda\Lib\site-packages\sklearn\base.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel(). return fit_method(estimator, *args, **kwargs)`

Absolute_error	None	100
----------------	------	-----



The screenshot shows a Jupyter Notebook interface with the following code and output:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[64]: regressor=RandomForestRegressor(n_estimators=100, criterion='absolute_error', random_state=0)
[65]: regressor.fit(x_train,y_train)
[66]: y_pred=regressor.predict(x_test)
[67]: #Evaluation Metrics
[68]: from sklearn.metrics import r2_score
[69]: r_score=r2_score(y_test,y_pred)
[70]: r_score
[70]: 0.9459897466049424
```

A warning message is displayed: `C:\Anaconda\Lib\site-packages\sklearn\base.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel(). return fit_method(estimator, *args, **kwargs)`

Absolute_error

sqrt

100

The screenshot shows a Jupyter Notebook interface with the following code and output:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[71]: regressor=RandomForestRegressor(n_estimators=100, criterion='absolute_error', random_state=0, max_features='sqrt')
[72]: regressor.fit(x_train,y_train)
[73]: y_pred=regressor.predict(x_test)
[74]: #Evaluation Metrics
[75]: from sklearn.metrics import r2_score
[76]: r_score=r2_score(y_test,y_pred)
[77]: r_score
[77]: 0.7857483352154211
```

The output shows the following values:

```
26 105733.54
34 96712.80
18 124266.90
```

A warning message is displayed: `C:\Anaconda\Lib\site-packages\sklearn\base.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().`

Absolute_error

Log2

100

The screenshot shows a Jupyter Notebook interface with the following code and output:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[78]: regressor=RandomForestRegressor(n_estimators=100, criterion='absolute_error', random_state=0, max_features='log2')
[79]: regressor.fit(x_train,y_train)
[80]: y_pred=regressor.predict(x_test)
[81]: #Evaluation Metrics
[82]: from sklearn.metrics import r2_score
[83]: r_score=r2_score(y_test,y_pred)
[84]: r_score
[84]: 0.7857483352154211
```

The output shows the following values:

```
26 105733.54
34 96712.80
18 124266.90
```

A warning message is displayed: `C:\Anaconda\Lib\site-packages\sklearn\base.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().`

Friedman_mse	sqrt	50
--------------	------	----

The screenshot shows a JupyterLab window titled "RandomForestRegression" with a last checkpoint 29 minutes ago. The left sidebar displays the Anaconda Toolbox with options for creating projects, notebooks, and managing snippets. The main code area contains the following Python code:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[85]: regressor=RandomForestRegressor(n_estimators=50, criterion='friedman_mse', random_state=0, max_features='sqrt')
[86]: regressor.fit(x_train,y_train)
[87]: y_pred=regressor.predict(x_test)
[88]: #Evaluation Metrics
[89]: from sklearn.metrics import r2_score
[90]: r_score=r2_score(y_test,y_pred)
[91]: r_score
[91]: 0.6889182139535486
```

A warning message is displayed: "DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().". A tooltip for the RandomForestRegressor object is visible, showing its parameters.

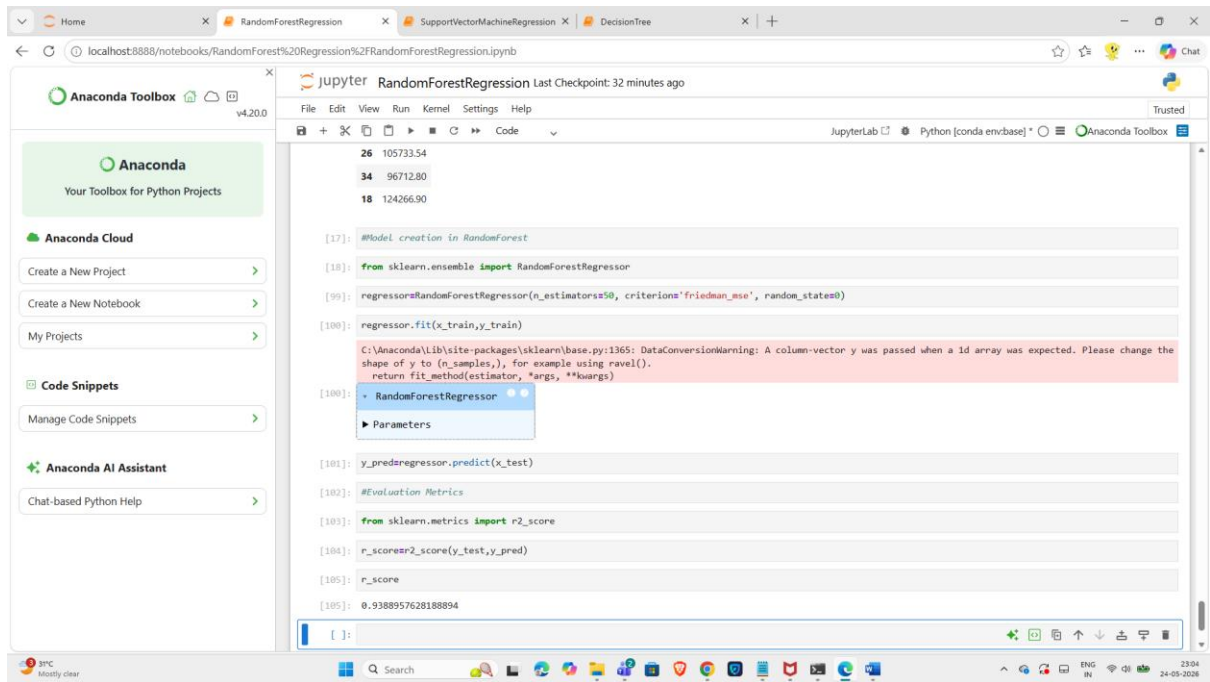
Friedman_mse	Log2	50
--------------	------	----

The screenshot shows a JupyterLab window titled "RandomForestRegression" with a last checkpoint 30 minutes ago. The left sidebar displays the Anaconda Toolbox with options for creating projects, notebooks, and managing snippets. The main code area contains the following Python code:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[92]: regressor=RandomForestRegressor(n_estimators=50, criterion='friedman_mse', random_state=0, max_features='log2')
[93]: regressor.fit(x_train,y_train)
[94]: y_pred=regressor.predict(x_test)
[95]: #Evaluation Metrics
[96]: from sklearn.metrics import r2_score
[97]: r_score=r2_score(y_test,y_pred)
[98]: r_score
[98]: 0.6889182139535486
```

A warning message is displayed: "DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().". A tooltip for the RandomForestRegressor object is visible, showing its parameters.

Friedman_mse	None	50
--------------	------	----

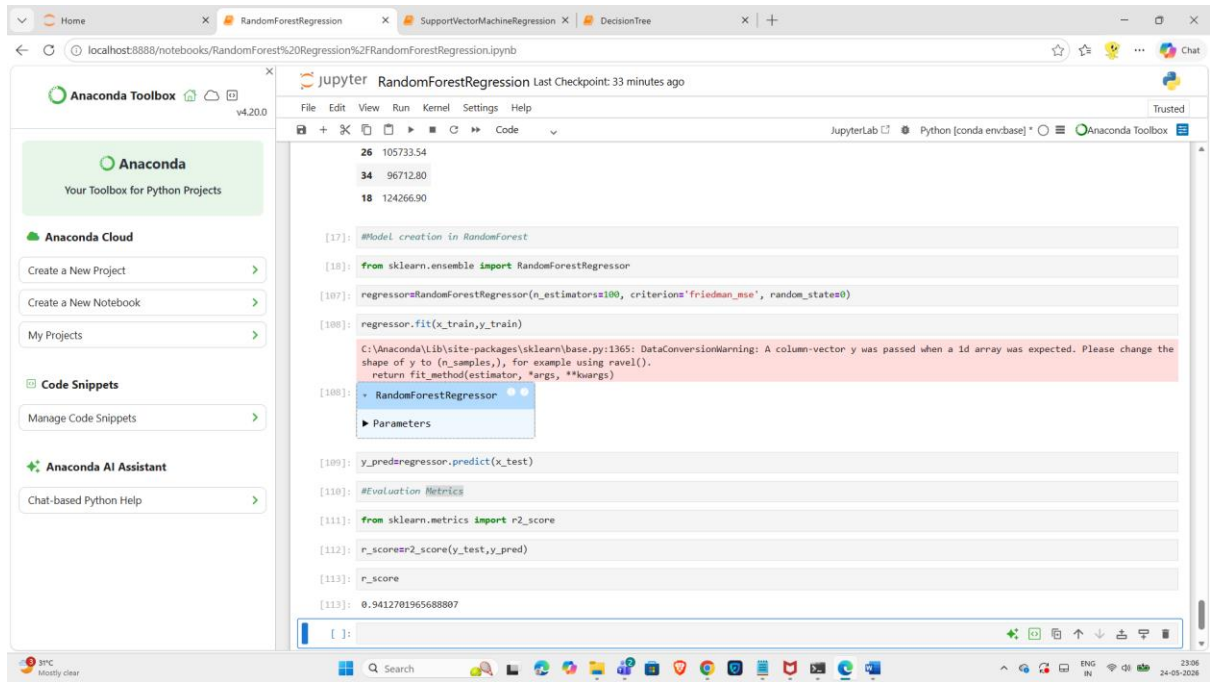


The screenshot shows a Jupyter Notebook interface with the following code and output:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[99]: regressor=RandomForestRegressor(n_estimators=50, criterion='friedman_mse', random_states=0)
[100]: regressor.fit(x_train,y_train)
[101]: y_pred=regressor.predict(x_test)
[102]: #Evaluation Metrics
[103]: from sklearn.metrics import r2_score
[104]: r_score=r2_score(y_test,y_pred)
[105]: r_score
[105]: 0.9388957628188894
```

A warning message is displayed: `C:\Anaconda\Lib\site-packages\sklearn\base.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel(). return fit_method(estimator, *args, **kwargs)`

Friedman_mse	None	100
--------------	------	-----



The screenshot shows a Jupyter Notebook interface with the following code and output:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[107]: regressor=RandomForestRegressor(n_estimators=100, criterion='friedman_mse', random_states=0)
[108]: regressor.fit(x_train,y_train)
[109]: y_pred=regressor.predict(x_test)
[110]: #Evaluation Metrics
[111]: from sklearn.metrics import r2_score
[112]: r_score=r2_score(y_test,y_pred)
[113]: r_score
[113]: 0.9412781965688887
```

A warning message is displayed: `C:\Anaconda\Lib\site-packages\sklearn\base.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel(). return fit_method(estimator, *args, **kwargs)`

Friedman_mse	sqrt	100
--------------	------	-----

The screenshot shows a Jupyter Notebook interface with the following code and output:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[114]: regressor=RandomForestRegressor(n_estimators=100, criterion='friedman_mse', random_states=0, max_features='sqrt')
[115]: regressor.fit(x_train,y_train)
C:\Anaconda\Lib\site-packages\sklearn\base.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
return fit_method(estimator, *args, **kwargs)
[115]: RandomForestRegressor
Parameters
[116]: y_pred=regressor.predict(x_test)
[117]: #Evaluation Metrics
[118]: from sklearn.metrics import r2_score
[119]: r_score=r2_score(y_test,y_pred)
[120]: r_score
[120]: 0.7608592212109025
```

Friedman_mse	Log2	100
--------------	------	-----

The screenshot shows a Jupyter Notebook interface with the following code and output:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[121]: regressor=RandomForestRegressor(n_estimators=100, criterion='friedman_mse', random_states=0, max_features='log2')
[122]: regressor.fit(x_train,y_train)
C:\Anaconda\Lib\site-packages\sklearn\base.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
return fit_method(estimator, *args, **kwargs)
[122]: RandomForestRegressor
Parameters
[123]: y_pred=regressor.predict(x_test)
[124]: #Evaluation Metrics
[125]: from sklearn.metrics import r2_score
[126]: r_score=r2_score(y_test,y_pred)
[127]: r_score
[127]: 0.7608592212109025
```


poisson	sqrt	50
---------	------	----

The screenshot shows a Jupyter Notebook titled "RandomForestRegression" with the following code and output:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[128]: regressor=RandomForestRegressor(n_estimators=50, criterion='poisson', random_state=0, max_features='sqrt')
[129]: regressor.fit(x_train,y_train)
[130]: y_pred=regressor.predict(x_test)
[131]: #Evaluation Metrics
[132]: from sklearn.metrics import r2_score
[133]: r_score=r2_score(y_test,y_pred)
[134]: r_score
[134]: 0.728862466757838
```

A warning message is displayed: "DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel(). return fit_method(estimator, *args, **kwargs)".

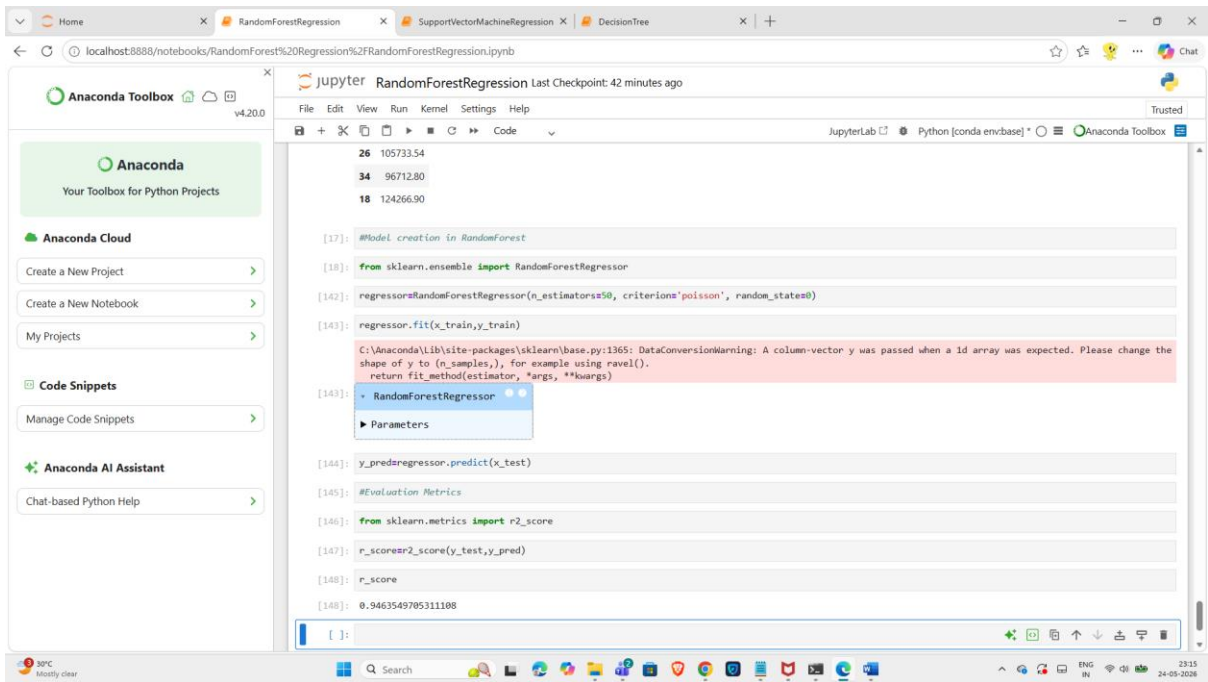
poisson	Log2	50
---------	------	----

The screenshot shows a Jupyter Notebook titled "RandomForestRegression" with the following code and output:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[135]: regressor=RandomForestRegressor(n_estimators=50, criterion='poisson', random_state=0, max_features='log2')
[136]: regressor.fit(x_train,y_train)
[137]: y_pred=regressor.predict(x_test)
[138]: #Evaluation Metrics
[139]: from sklearn.metrics import r2_score
[140]: r_score=r2_score(y_test,y_pred)
[141]: r_score
[141]: 0.728862466757838
```

A warning message is displayed: "DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel(). return fit_method(estimator, *args, **kwargs)".

poisson	None	50
---------	------	----



The screenshot shows a Jupyter Notebook titled "RandomForestRegression" with the following code and output:

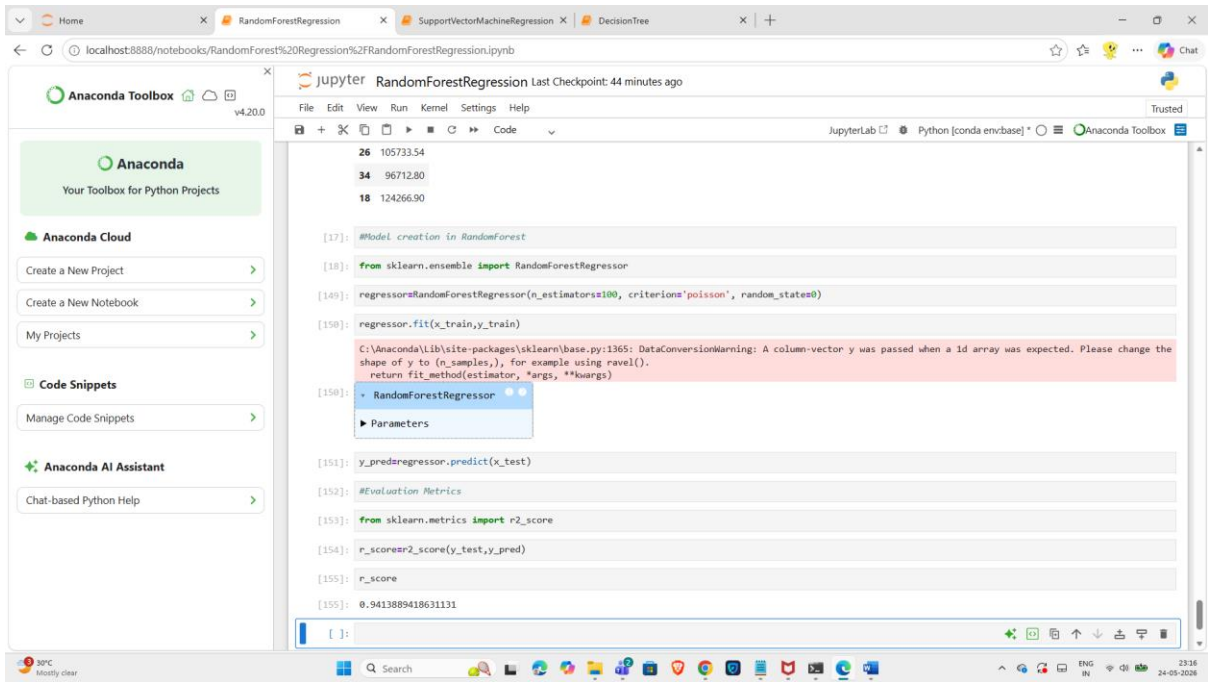
```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[142]: regressor=RandomForestRegressor(n_estimators=50, criterion='poisson', random_state=0)
[143]: regressor.fit(x_train,y_train)
[144]: y_pred=regressor.predict(x_test)
[145]: #Evaluation Metrics
[146]: from sklearn.metrics import r2_score
[147]: r_score=r2_score(y_test,y_pred)
[148]: r_score
[148]: 0.9463549705311108
```

The output of the fit method is a `RandomForestRegressor` object with the following parameters:

- `n_estimators`: 50
- `criterion`: poisson
- `random_state`: 0

The notebook also shows the prediction of the test set using the trained model, resulting in an `r2_score` of 0.9463549705311108.

poisson	None	100
---------	------	-----



The screenshot shows a Jupyter Notebook titled "RandomForestRegression" with the following code and output:

```
[17]: #Model creation in RandomForest
[18]: from sklearn.ensemble import RandomForestRegressor
[149]: regressor=RandomForestRegressor(n_estimators=100, criterion='poisson', random_state=0)
[150]: regressor.fit(x_train,y_train)
[151]: y_pred=regressor.predict(x_test)
[152]: #Evaluation Metrics
[153]: from sklearn.metrics import r2_score
[154]: r_score=r2_score(y_test,y_pred)
[155]: r_score
[155]: 0.9413889418631131
```

The output of the fit method is a `RandomForestRegressor` object with the following parameters:

- `n_estimators`: 100
- `criterion`: poisson
- `random_state`: 0

The notebook also shows the prediction of the test set using the trained model, resulting in an `r2_score` of 0.9413889418631131.

poisson	sqrt	100
---------	------	-----

Home RandomForestRegression SupportVectorMachineRegression DecisionTree

localhost:8888/notebooks/RandomForest%20Regression%2FRandomForestRegression.ipynb

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JupyterLab Python [conda envbase] Anaconda Toolbox

```
26 105733.54
34 96712.80
18 124266.90

[17]: #Model creation in RandomForest

[18]: from sklearn.ensemble import RandomForestRegressor

[156]: regressor=RandomForestRegressor(n_estimators=100, criterion='poisson', random_state=0, max_features='sqrt')

[157]: regressor.fit(x_train,y_train)

C:\Anaconda\Lib\site-packages\sklearn\base.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
return fit_method(estimator, *args, **kwargs)

[157]: RandomForestRegressor
Parameters

[158]: y_pred=regressor.predict(x_test)

[159]: #Evaluation Metrics

[160]: from sklearn.metrics import r2_score

[161]: r_score=r2_score(y_test,y_pred)

[162]: r_score

[162]: 0.7717642068103981
```

30°C Mostly clear 2318 24-05-2028

poisson	Log2	100
---------	------	-----

Home RandomForestRegression SupportVectorMachineRegression DecisionTree

localhost:8888/notebooks/RandomForest%20Regression%2FRandomForestRegression.ipynb

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JupyterLab Python [conda envbase] Anaconda Toolbox

```
26 105733.54
34 96712.80
18 124266.90

[17]: #Model creation in RandomForest

[18]: from sklearn.ensemble import RandomForestRegressor

[163]: regressor=RandomForestRegressor(n_estimators=100, criterion='poisson', random_state=0, max_features='log2')

[164]: regressor.fit(x_train,y_train)

C:\Anaconda\Lib\site-packages\sklearn\base.py:1365: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
return fit_method(estimator, *args, **kwargs)

[164]: RandomForestRegressor
Parameters

[165]: y_pred=regressor.predict(x_test)

[166]: #Evaluation Metrics

[167]: from sklearn.metrics import r2_score

[168]: r_score=r2_score(y_test,y_pred)

[169]: r_score

[169]: 0.7717642068103981
```

30°C Mostly clear 2319 24-05-2028